

AgiBot X2 Ground Recovery

Simulation, reinforcement learning and ROS 2 integration | 24 September 2026

Result. The trained PPO policy recovered from a true supine start in all five fixed-seed, 10-second Isaac Lab episodes. It remained upright on both feet without other-body ground support at the end of each episode. The ROS 2 interface ran the same simulator-backed policy and passed acceptance, busy-request, telemetry and timeout checks.

Robot and simulation

Item	Implementation
Model	Official AgiBot X2 Ultra v1.3.0 URDF, pinned upstream revision 60c5de5; locally imported to USD.
Physics	Isaac Sim 6.0.1 / Isaac Lab 3.0.0-beta2.patch1 / PhysX; floating base, 200 x 200 m flat collision floor, self-collision.
Robot constraints	31 movable joints and 32 monitored rigid bodies; original geometry, inertia, joint axes and actuator effort/speed limits retained. Position targets use a 98% soft-limit margin.
Reset and rate	Supine at 0.190 m pelvis height, zero velocity and small seeded perturbations; no floor intersection. Physics 200 Hz, policy 50 Hz; 10 s or 500 policy steps.

The official source and import procedure are in `scripts/fetch_agibot_model.sh` and `scripts/import_x2_isaac.sh`. The imported USD remains local; it is regenerated from the pinned URDF rather than committed.

Observation and action contract

Policy input (122 values)	Count
Pelvis height; body-frame linear and angular velocity; projected gravity	1 + 3 + 3 + 3
Joint position and velocity	31 + 31
Left/right foot contact and whole-body ground-contact bits	2 + 32
Previous two policy actions	16

Eight symmetric action channels command hip pitch, knee, ankle pitch, shoulder pitch, elbow, waist pitch, ankle roll and hip roll. They vary targets for 15 joints. The other 16 retain default position targets and remain active in physics and telemetry. Targets are mapped with $\text{centre} + \text{span} * \tanh(\text{action})$, then clamped to imported soft joint limits. The policy outputs position targets, not direct torques. Actor and critic use separate normalized 512/256/128 ELU MLPs.

Control smoothness

Small arm oscillations can remain even when recovery succeeds: the strict check constrains base speeds and support, and the additional arm check bounds position error, not arm velocity. Playback uses deterministic mean actions; it does not sample the training Gaussian. Action-change and torque penalties are weak, with no explicit arm-velocity objective. Feedback targets and joint-drive dynamics can sustain small corrections; their individual contributions have not been isolated. Further arm-velocity regularization or actuator tuning would require a separate evaluated experiment.

Reward design and PPO experiment

The reward combines rise progress, supported upright stance and control regularization. The signed-height term prevents a high inverted bridge from being rewarded. Non-foot contact is allowed during the push-off phase, then discouraged near standing; it is excluded by the final success check.

Term	Weight	Purpose / Implementation
Signed pelvis height	40	$\text{clip}(z/0.68) * \text{clip}((1-g_z)/2)$; height only counts when orientation is compatible with recovery.
Head height; upright	5; 5	Head-height progress and $\exp(-g_z)$ orientation reward.
Both feet; other support	5; -1	Reward current two-foot ground contact; penalize non-foot support as the pelvis rises.
Balance; strict stance	10; 20	Reward low speed near upright and the complete two-foot, unsupported success predicate.
Relaxed arms	40	Once upright on two feet without other support, favor shoulder pitch 0 and elbow -0.15 rad.
Action change; torque	-0.005; -1e-6	Penalize rapid target changes and squared actuator effort.
Joint limit; safety failure	-1; -10	Penalize soft-limit violations and non-timeout safety terminations.

The height/upright/contact objectives draw on HumanUP [1]; HoST [2] motivates a separate objective after standing; FRASA [3] motivates the compact symmetric action space. The X2 weights, numerical thresholds and gate formulas were selected for this robot and task. They are not a reproduction of those papers.

Training configuration

RSL-RL PPO used 3,000 parallel environments and 32 steps per environment, with clip 0.2, discount 0.99, GAE lambda 0.95, five epochs and four minibatches. The final 51-update, supine-only stage resumed from the supplied iteration-400 parent checkpoint with seed 47, fixed learning rate 1e-4, initial action standard deviation 0.10 and entropy coefficient 0.001. The selected checkpoint is reports/checkpoints/x2_relaxed_v4_model450.pt. Earlier exploration used auxiliary upright-root reset poses; the final training stage, evaluation and ROS runtime used only true-supine resets.

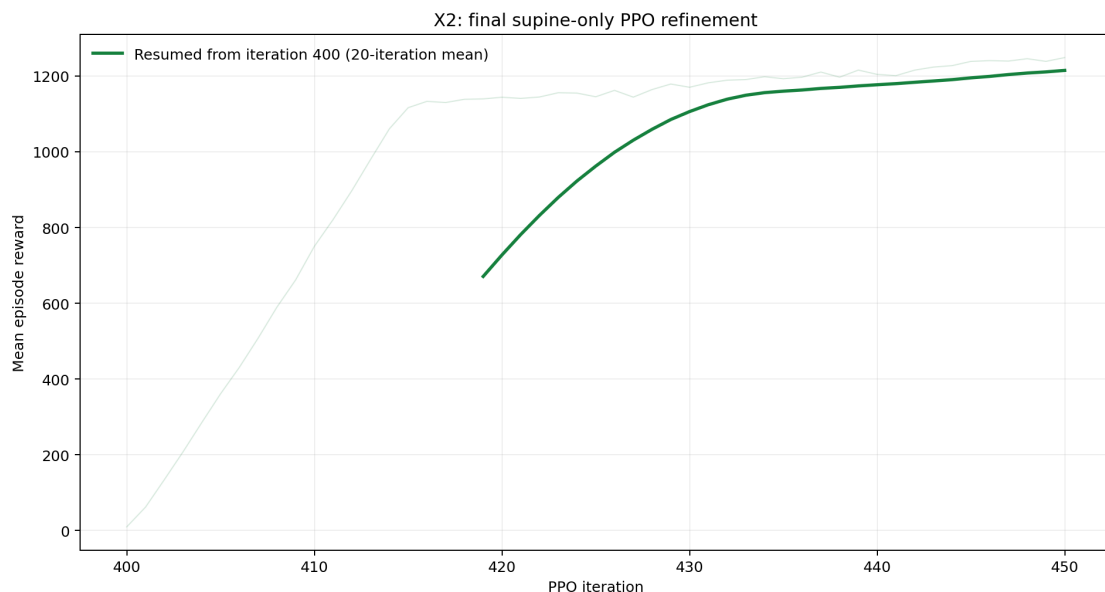


Figure 1. Final-stage mean episode reward; solid line is the 20-iteration moving mean. The curve shows learning progress; the five-episode physical success check is reported separately on page 3. Source: reports/relaxed_v4_training_reward.png.

Five-episode validation

Each run started supine with the same final policy and a fixed seed. A recovery was counted only after all conditions below held continuously for at least 0.5 s. The evaluator ran all 500 policy steps and checked the terminal state before reset, so a brief upright crossing was not counted as sustained recovery.

Check	Threshold
Pelvis height	At least 0.58 m.
Upright orientation	$ \text{projected_gravity_xy} \leq 0.15$ and $g_z \leq -0.98$.
Stable base	Linear speed at most 0.25 m/s; angular speed at most 0.35 rad/s.
Support	At least 15 N current ground force on each foot; less than 15 N on each other rigid body.
Additional posture report	Shoulder-pitch and elbow errors at most 0.30 rad while strict stance holds; assessed through the final two seconds.

Measured outcome

Seed	Recovery	Strict stance at end	Strict + relaxed arms at end	Failure
101	Yes	8.94 s	8.86 s	None
102	Yes	8.54 s	8.54 s	None
103	Yes	8.64 s	8.18 s	None
104	Yes	8.74 s	8.48 s	None
105	Yes	8.88 s	7.78 s	None

5/5 successful recoveries. All five episodes reached strict unsupported two-foot stance, retained it at episode end, and met the added relaxed-arm check through the final two seconds. No submitted evaluation episode ended by safety termination; all failure-reason fields are empty. The strict end-of-episode duration ranged from 8.54 to 8.94 s.

Source: `reports/relaxed_v4_evaluation.json`. A visual example is `reports/gifs_relaxed_v4/x2_final_policy_attempt.gif`. The selected checkpoint and reward CSV/PNG are under `reports/`.

Failure analysis and scope

Earlier height-only training found a high but inverted bridge supported by other body parts. Signed upright height and an explicit two-foot/no-other-support criterion removed that shortcut. A later policy stood but held its arms forward; the gated upper-body term improved final posture. These are development observations, not failures in the submitted five-seed evaluation. The result is for nominal flat-floor simulation. Hardware recovery, terrain variation, sensor noise, latency and broad initial-pose robustness were not validated.

Historical changes and their evidence are retained in `docs/development_history.md`. The available parent checkpoint reproduces the final 51-update stage; the complete exploratory lineage is documented but not claimed to be bit-for-bit reproducible.

Evaluation seeds

The assignment requires five simulation episodes; it does not prescribe seed numbers or a different task for each episode. We chose seeds 101-105 to reproduce five small variations of the same supine start. Root-position offsets are $X/Y \pm 4$ cm and $Z \pm 1$ mm; rotation offsets are roll/pitch ± 0.005 rad and yaw ± 0.02 rad. All runs use the same checkpoint, neutral joint angles and zero initial velocity. Seed 47 belongs to training, not this evaluation.

ROS 2 interface and reproduction

The Python package `src/x2_recovery_ros` builds with `colcon` and launches the recovery and telemetry nodes together. A local Unix socket connects ROS 2 Humble (Python 3.10) to the Isaac policy server (Python 3.12). Published joint positions come from the running simulator.

Interface	ROS 2 type	Behavior
<code>/x2/start_recovery</code>	<code>std_srvs/srv/Trigger</code>	Returns acceptance before execution; rejects a second request while pending or running.
<code>/x2/recovery_status</code>	<code>std_msgs/msg/String</code>	Publishes IDLE, RUNNING, SUCCEEDED or FAILED.
<code>/x2/joint_states</code>	<code>sensor_msgs/msg/JointState</code>	Publishes timestamps and all 31 measured joint names/positions while running.

The telemetry node logs status and `left_knee_joint` at 1 Hz. The recovery timeout is configurable. Recorded fresh-build and real simulator-backed outcomes are in `reports/ros_fresh_build_v4.txt` and `reports/ros_isaac_relaxed_v4_validation.txt`.

Recorded trial	Observed outcome
Normal CLI request	Accepted; RUNNING to SUCCEEDED; 100 live joint-state samples.
Concurrent request	One accepted; the other rejected as busy; 85 joint-state samples.
0.2 s timeout	Accepted; RUNNING to FAILED after 10 simulator steps/samples.

Commands from the repository root

Install the pinned simulator and ROS dependencies as described in `README.md` and `docs/commands.md`. The supplied checkpoint supports playback and evaluation without retraining. Choose a seed with `--seeds`; `--viz kit` opens the graphical viewer.

```
export ISAAC_PYTHON=/path/to/isaac/environment/bin/python
./scripts/fetch_agibot_model.sh
./scripts/import_x2_isaac.sh
CKPT=reports/checkpoints/x2_relaxed_v4_model450.pt
./scripts/play_isaac.sh "$CKPT" --environment relaxed_v4 \
  --device cuda:0 --viz kit --seeds 101 --output /tmp/x2_play.json
./scripts/evaluate_isaac.sh "$CKPT" --environment relaxed_v4 \
  --device cuda:0 --output /tmp/x2_evaluation.json
```

For ROS, build with `./scripts/build_ros.sh`, start the Isaac policy server in one terminal, then launch both ROS nodes in a separately prepared ROS terminal. Call `/x2/start_recovery` and watch the status and joint topics before the request. Exact multi-terminal preparation, busy rejection and timeout steps are in `docs/ros_live_validation.md`.

New training runs write their own timestamped checkpoint, reward PNG/CSV and manifest under `logs/rs1_rl/`. They do not overwrite the submitted artifacts in `reports/`.

Evidence and references

Repository evidence: `README.md` (setup and design); `docs/commands.md` (ordered commands); `docs/validation.md` (outcomes); `docs/parameter_provenance.md` (design-choice sources); `reports/` (checkpoint, reward curve, five-episode JSON and ROS logs). Development history remains in Git and `docs/development_history.md`. Tested host: Ubuntu 22.04.5 with one RTX 5080 Laptop GPU (16 GB). Vendor collision definitions are retained; mesh colliders use the converter's convex-hull approximation.

- [1] He et al., "Learning Getting-Up Policies for Real-World Humanoid Robots," RSS 2025. doi:10.15607/RSS.2025.XXI.063.
- [2] Huang et al., "Learning Humanoid Standing-up Control across Diverse Postures," RSS 2025. doi:10.15607/RSS.2025.XXI.064.
- [3] Gaspard et al., "FRASA." arXiv:2410.08655.
- [4] Schulman et al., "Proximal Policy Optimization Algorithms." arXiv:1707.06347.

Playback preview

Add `--start-delay 2` to graphical playback to show the frozen supine reset pose for two wall-clock seconds before each rollout. The viewer updates while physics and the episode clock remain at reset. Use 0 for immediate playback; headless evaluation has no preview.